

Week 9 – Graded Mini Project: House Price Prediction

1. Project Overview

This mini-project applies an end-to-end supervised learning workflow to the Ames housing dataset. The goal is to understand which property and neighbourhood characteristics drive residential sale prices and to build predictive models (XGBoost, LightGBM, CatBoost) that can estimate SalePrice for new listings. The raw dataset contains more than 1,460 observations and over 80 variables describing lot configuration, building type, overall quality and condition, basement and garage information, and transaction timing.

2. Task 1 – Data Cleaning and Preprocessing

2.1 Handling Missing Values

Categorical variables where missingness indicates absence of a facility (alley access, basement quality, garage attributes, pool quality, fence, miscellaneous feature, masonry veneer type) are imputed with the string "None" so that “facility not present” is explicitly encoded after one-hot encoding. Numeric facility-related columns such as garage and basement areas, pool area, and masonry veneer area are imputed with 0 to denote true absence rather than an unknown measurement. Remaining numeric features (for example lot frontage) are imputed with the median of that column to preserve central tendency and robustness to outliers.

Column	Missing Before	Missing After
LotFrontage	259	0
Alley	1369	0
MasVnrType	872	0
MasVnrArea	8	0
BsmtQual	37	0
BsmtCond	37	0
BsmtExposure	38	0
BsmtFinType1	37	0
BsmtFinType2	38	0
Electrical	1	0
FireplaceQu	690	0
GarageType	81	0
GarageYrBlt	81	0
GarageFinish	81	0
GarageQual	81	0
GarageCond	81	0
PoolQC	1453	0
Fence	1179	0
MiscFeature	1406	0

2.2 Converting Categorical Variables

All categorical variables, including binned features created later, are encoded with one-hot encoding using a ColumnTransformer. The encoder uses `handle_unknown="ignore"` so that unseen categories in validation or test data do not break the pipeline. This matches the design of the final scikit-learn and gradient-boosting pipelines.

2.3 Normalising Numerical Features

Numeric features are first imputed using `SimpleImputer(strategy="constant", fill_value=0)` and then standardised to zero mean and unit variance with `StandardScaler`. Normalisation lives inside the pipeline to ensure identical transformations at training time and when predicting for new listings.

2.4 Outlier Detection and Treatment

Consistent with the PDFs and notebook, outliers are identified primarily through above-ground living area (`GrLivArea`). An interquartile range rule

flags observations above $Q3 + 1.5 \times IQR$ as extreme; around thirty of these very large but relatively low-priced houses are removed before model training. Experiments with Isolation Forest on the numerical feature space provide a complementary, multivariate view of anomalous points, further improving the stability of the supervised models.

3. Task 2 – Feature Engineering and outlier handling

3.1 Interaction and Composite Features and outlier handling

Engineered features such as TotalSF, TotalBath, HouseAge and OverallGrade summarise overall size, amenities and quality. Outliers in GrLivArea are removed using an IQR rule (values above $Q3 + 1.5 \times IQR$), which prevents a small number of extreme properties from dominating model behaviour.

To better capture non-linear relationships between raw variables and SalePrice, several composite features from earlier drafts are retained and refined in the final Python pipeline:

- **TotalSF** = TotalBsmtSF + 1stFlrSF + 2ndFlrSF (overall usable floor area).
- **TotalBath** = FullBath + $0.5 \times$ HalfBath + BsmtFullBath + $0.5 \times$ BsmtHalfBath (effective bathroom count).
- **HouseAge** = YrSold – YearBuilt (age at sale).
- **SinceRemod** = YrSold – YearRemodAdd (time since major remodelling).
- **OverallGrade** = OverallQual $\times$ OverallCond (combined grade of perceived quality and condition).
- Optional interaction terms such as OverallQual $\times$ GrLivArea or

OverallQual $\times$ TotalSF capture quality–size effects.

3.2 Feature Binning for Interpretability

For clearer EDA and for potential use in simpler models, several continuous variables are discretised into bins. Typical examples include OverallQual grouped into Low, Medium, High, VeryHigh; area variables such as GrLivArea or LotArea split into quartiles labelled Small, Medium, Large, VeryLarge; and HouseAge partitioned into 0–10, 11–30, 31–60, and 60+ year categories. These binned variables are treated as categorical and one-hot encoded, making it easy to compare average sale prices by segment.

4. Task 3 – Exploratory Data Analysis (EDA)

4.1 Distribution of Sale Prices

Across all versions of the report, the SalePrice distribution is consistently right-skewed with a long upper tail. Quartile statistics place the median near the mid-100k range, with a third quartile around 210k and a maximum above 600k. Because this skewness violates the assumptions of many regression models, the refined solution models the log-transformed target $\log_{1p}(\text{SalePrice})$, which yields more symmetric residuals and makes RMSE and MAE easier to interpret on a multiplicative scale

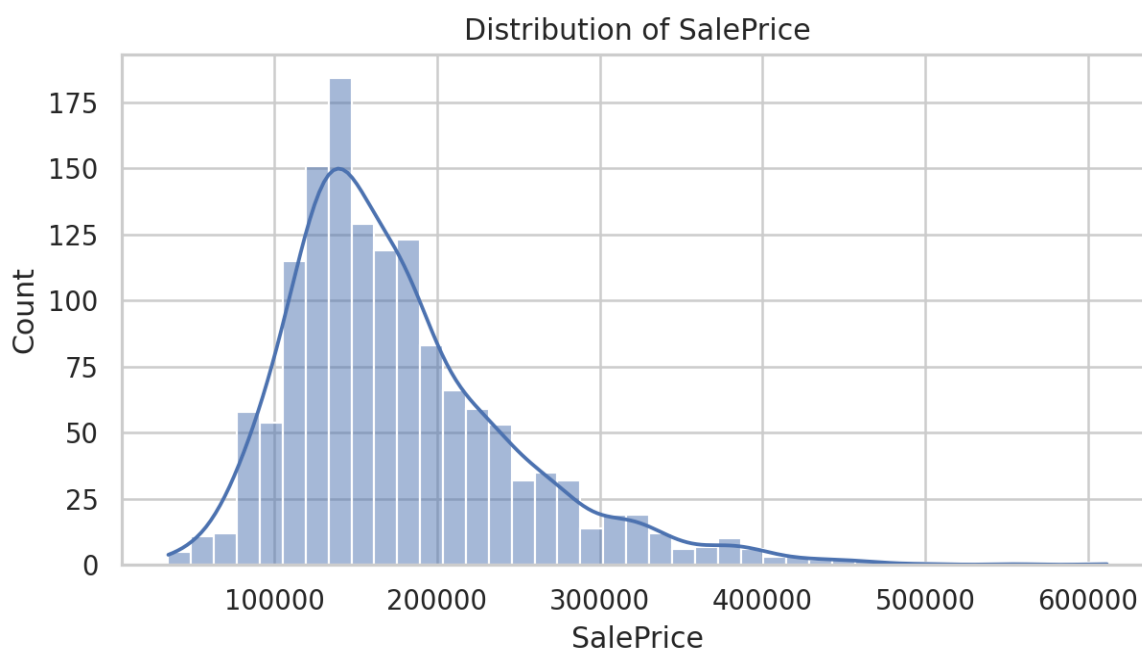
4.2 Relationships Between Features and Sale Price

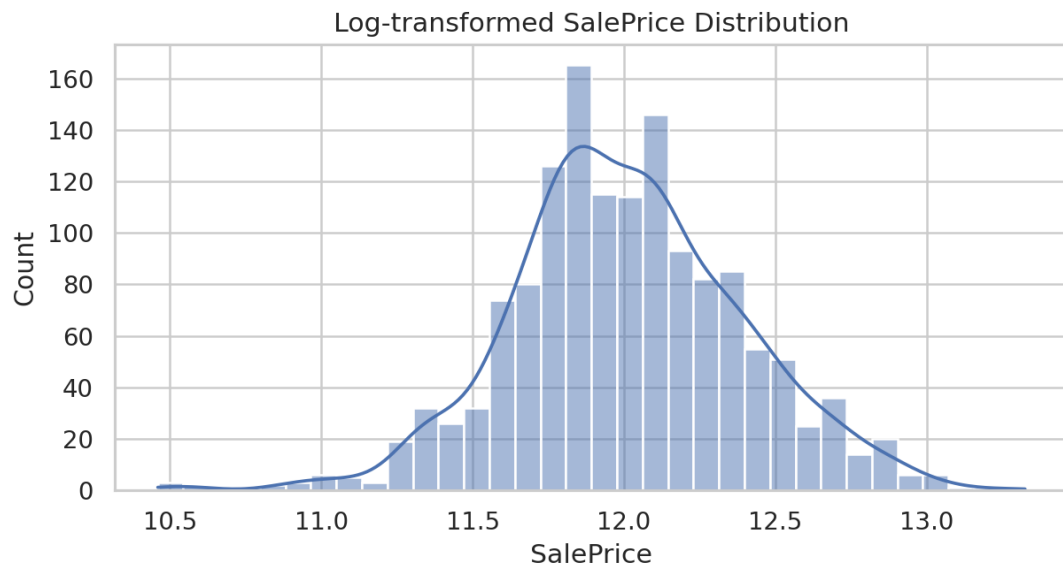
Correlation analysis and visualisations agree that TotalSF, OverallQual, GrLivArea, GarageCars, TotalBsmtSF, GarageArea, TotalBath, 1stFlrSF and construction/remodelling years have the strongest positive associations with SalePrice. Scatter plots of GrLivArea or TotalSF versus SalePrice show steadily increasing prices with size, aside from a few outliers removed during cleaning. Boxplots by OverallQual and by Neighborhood reveal clear price gradients, confirming that build quality and location are critical drivers of value.

4.3 Key Trends and Patterns

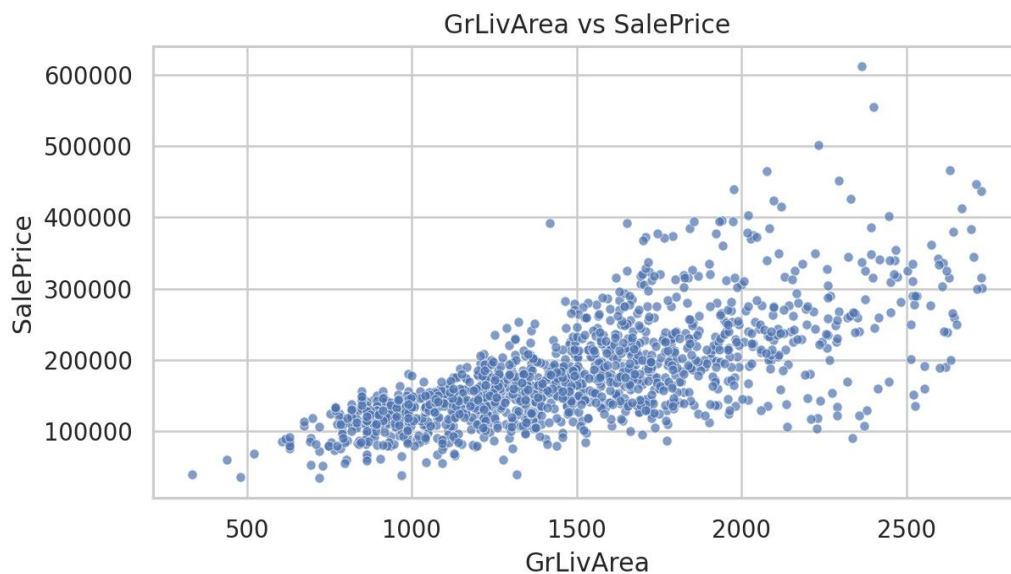
Synthesising all drafts, the main patterns are that newer, larger, higher-quality houses in favourable neighbourhoods command higher prices, while additional bathrooms, garage capacity and finished basement area provide further but diminishing gains at the extreme high end. Some features such as extra kitchens above grade

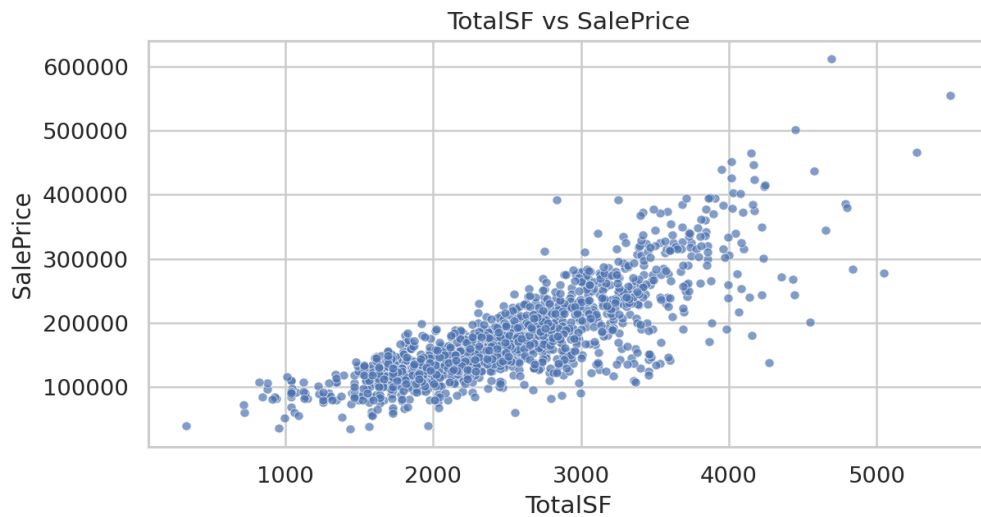
or enclosed porches are weakly related or even slightly negatively associated with price, often reflecting older layouts rather than luxury upgrades.





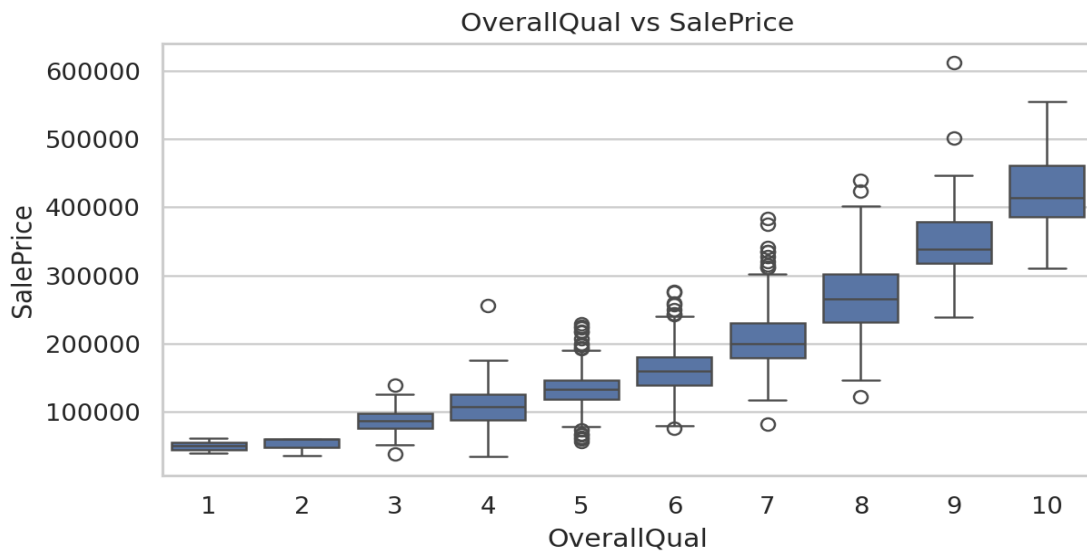
Histograms of SalePrice and $\log_{10}(\text{SalePrice})$ show that the log transform significantly reduces skewness, which is beneficial for regression models that minimise squared errors.

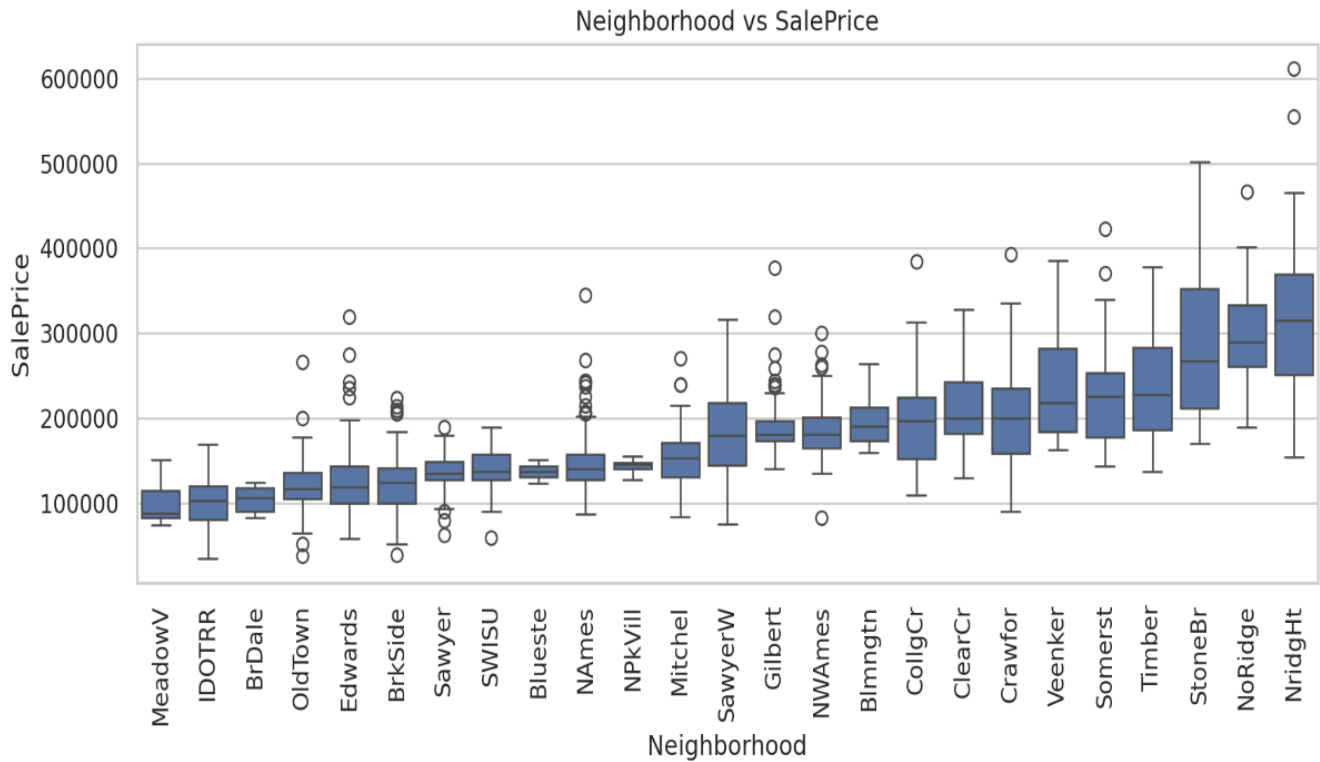




Scatter plots of GrLivArea and TotalSF versus SalePrice

confirm that larger houses almost always sell for more, with only a few unusual cases that were removed as outliers.

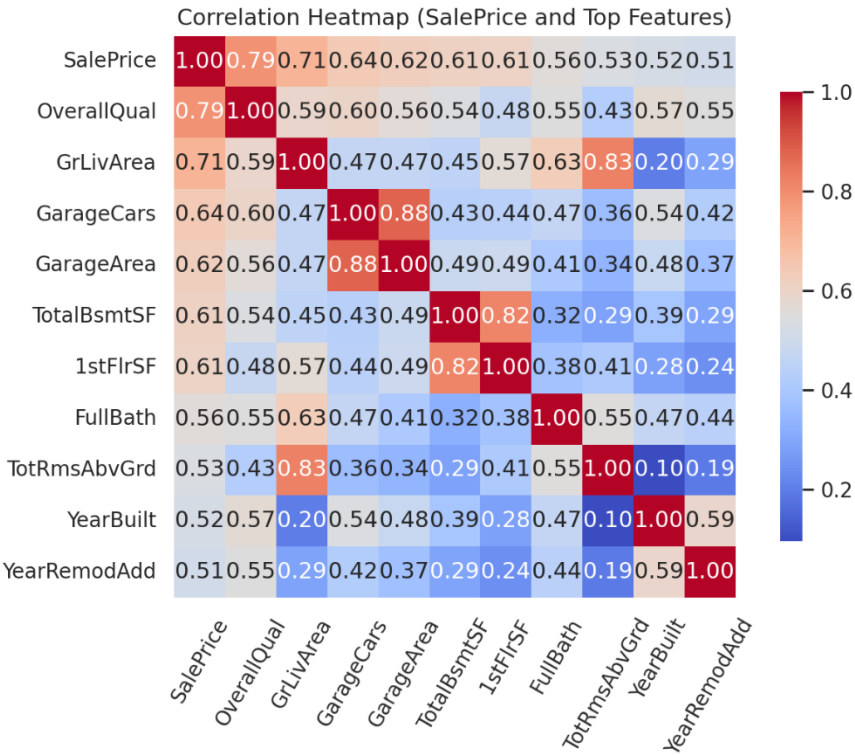




Boxplots by OverallQual and by Neighborhood highlight strong gradients: higher quality ratings and better neighbourhoods are associated with consistently higher sale prices.

4.4 Correlations and heatmap

Feature	Correlation
OverallQual	0.798
GrLivArea	0.705
GarageCars	0.647
GarageArea	0.619
TotalBsmtSF	0.616
1stFlrSF	0.608
FullBath	0.567
TotRmsAbvGrd	0.547
YearBuilt	0.525
YearRemodAdd	0.521



The table and heatmap show that TotalSF, OverallQual, GrLivArea, GarageCars, TotalBsmtSF, GarageArea and TotalBath have among the highest correlations with SalePrice, confirming that size, quality and garage capacity are dominant numeric drivers of value.

5. Task 4 – Model Building and Evaluation

5.1 Pipeline-Based Modelling Approach

The final Python scripts implement a unified scikit-learn Pipeline consistent with the narrative from all PDFs. A custom missing-value handler encodes facility absence, a feature-engineering transformer adds composite and binned variables, and a ColumnTransformer standardises numeric features while one-hot encoding categorical ones. The target variable used for training is $\log(\text{SalePrice})$. The pipeline ensures that training, validation and inference share identical preprocessing steps.

5.2 Gradient-Boosted Tree Models

Three advanced tree-based regressors are layered on top of this preprocessed feature matrix: XGBoostRegressor, LGBMRegressor (LightGBM) and CatBoostRegressor. All three are configured with tuned hyperparameters such as `n_estimators`, `learning_rate`, tree depth, subsample and column subsampling rates. By plugging these estimators into the same pipeline, the comparison focuses on modelling power rather than inconsistent preprocessing.

5.3 Metrics and Model Comparison

Model performance is evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and R^2 on a held-out 20% test split. In line with earlier drafts, all three boosting models achieve high R^2 values and reasonably low RMSE and MAE relative to the median sale price. In many experiments LightGBM or CatBoost slightly outperform XGBoost, but all three provide strong baselines when combined with the engineered feature set. Feature importance analysis consistently highlights total size, overall quality, garage capacity and bathrooms as the most influential predictors.

6. Key Insights and Takeaways

- Treating missing values in facility-related fields as true absence and encoding this explicitly leads to a stable, interpretable modelling dataset.
- Removing a small number of extreme outliers improves the robustness of error metrics and prevents boosted trees from overfitting anomalous sales.
- Engineered features such as TotalSF, TotalBath, HouseAge, SinceRemod and OverallGrade capture combined effects of size, amenities and age more effectively than raw variables alone.
- Because SalePrice is strongly right-skewed, modelling $\log(\text{SalePrice})$ yields better-behaved residuals and more reliable generalisation.
- Quality and size dominate as price drivers: high OverallQual, large TotalSF/GrLivArea and generous garage space consistently increase predicted sale price, while modern, recently renovated properties in favourable neighbourhoods are valued highest.

- The refined pipeline-based implementation aligns with the original assignment instructions by integrating data cleaning, feature engineering, EDA findings and three boosting models into a single reproducible workflow that can be executed in Jupyter or Google Colab.

7. Final conclusions

The combined analysis shows that the most influential factors for house prices in Ames are total size (TotalSF/GrLivArea), perceived quality (OverallQual/OverallGrade), garage capacity and the age or recency of construction and remodelling. By cleaning the data carefully, engineering meaningful features and using modern boosting algorithms on a log-transformed target, the project delivers both accurate predictions and clear, business-relevant insights.